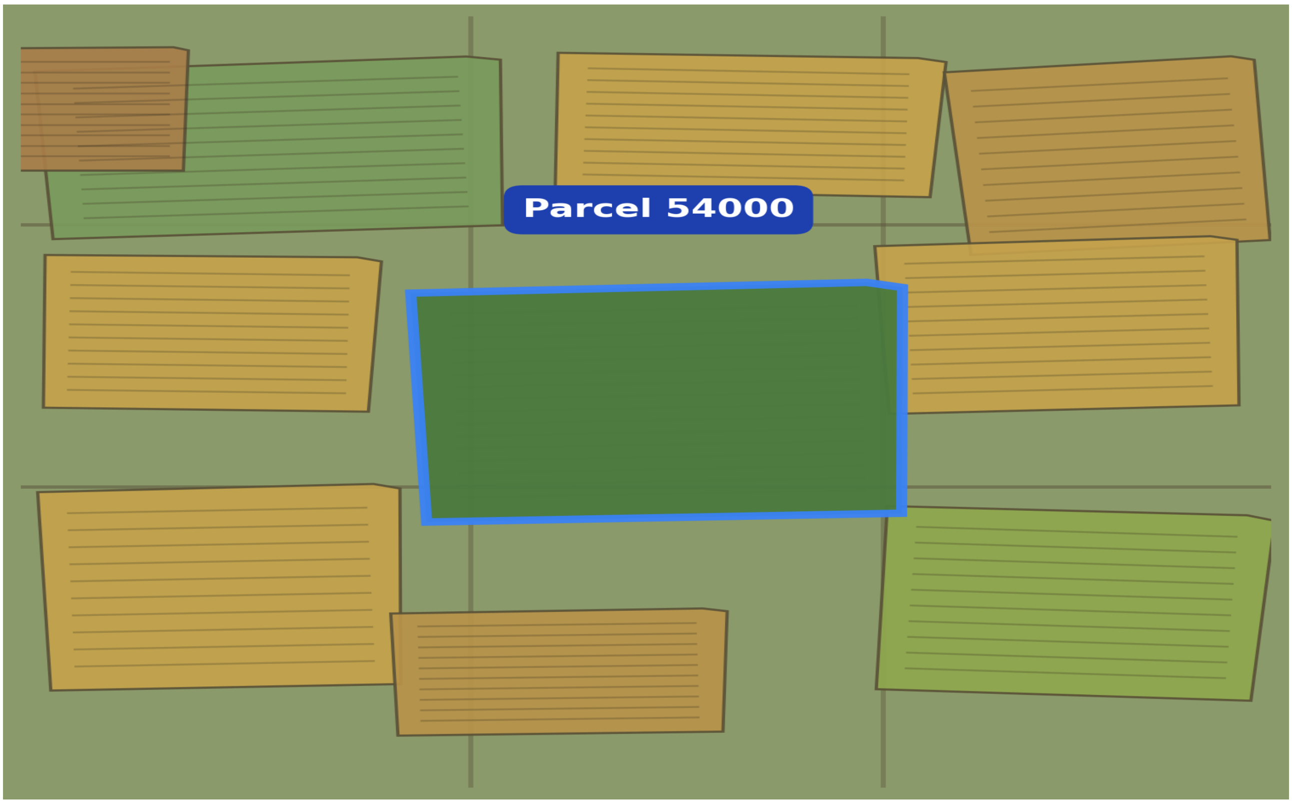
Cube Earth

SAMPLE REPORT

Field Intelligence Report

Parcel 54000 · Generated 27 July 2026 · LPIS 2024



Selected parcel (blue outline) among surrounding mapped fields, coloured by crop type (illustrative)

Crop Detection

SATELLITE ESTIMATE	LPIS DECLARATION (FARMER)
■ **Maize**	■ **Barley - Spring**
Confidence: **100%** ■ High	Status: ■ **Satellite estimate differs**

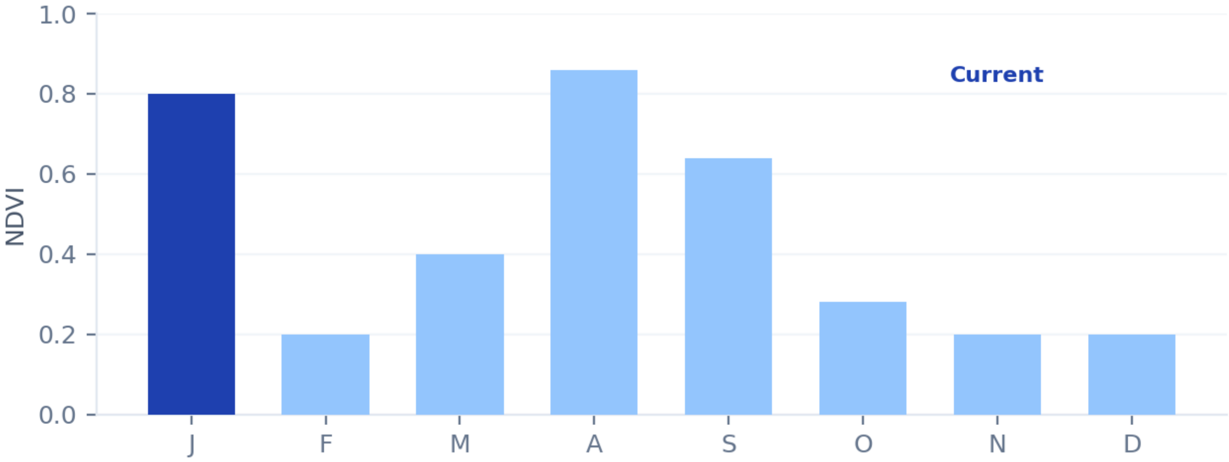
This difference is presented as evidence for review, not as a determination that the declaration is incorrect — see Methodology for how differences are calculated.

1.6 ha	**Silking**	**0.80**	**5–15 Oct**
FIELD AREA	GROWTH STAGE (4/7)	NDVI (VEGETATION)	HARVEST WINDOW

Growth Timeline

✓ Emergence	✓ Vegetative	✓ Tasselling	■ Silking	Grain Fill	Dough	Maturity

NDVI History (2025)



Satellite Observations

Metric	Value	Note
NDVI (Vegetation)	0.80	Dense canopy
Soil Moisture	Adequate	From weather-model estimate
Trafficability	Good	Model-derived, not field-verified
Cloud cover (7-day)	~45%	Affects imagery freshness

7-Day Weather Forecast

6 mm	9–22°C	20 km/h
EXPECTED RAIN	TEMPERATURE RANGE	WIND

AI Insights

■ High confidence difference from LPIS

Satellite strongly suggests Maize but the LPIS declaration is Barley - Spring. This parcel may warrant ground verification.

■ Healthy crop

Vegetation indices indicate good canopy development. No signs of stress detected from satellite data.

■ Next satellite pass

HLS revisit in approximately 4 days. Updated NDVI and vegetation maps will be available.

Parcel Information

Area	1.6 ha (~4 acres)	**Parcel ID**	54000
Centroid Latitude	53.1602°N	**Centroid Longitude**	7.0502°W
Source	LPIS 2024	**Data vintage**	2022–2025 (4 seasons)

Methodology & Disclaimer

How this report was produced

This crop prediction is generated using a CatBoost machine learning model trained on 61,566 Irish tillage parcels from the LPIS 2024 dataset, using HLS (Harmonized Landsat Sentinel-2) satellite imagery — NDVI, EVI, NDWI, NDRE and NDII vegetation indices — across four growing seasons (2022–2025). The model achieves 77.05% balanced accuracy nationally on held-out validation data. Confidence scores are calibrated using isotonic regression so that the displayed percentage reflects genuine prediction reliability.

Reading confidence scores

■ **High (80–100%)**	Prediction is usually reliable.
■ **Medium (50–79%)**	Similar crops may also fit the satellite signature.
■ **Low (<50%)**	Use as a prompt for further review, not as a conclusion.

Known limitations

• Spring cereals (Barley, Oats, Wheat) are more easily confused with each other than with Maize or winter crops.
• Satellite observations are affected by cloud cover; confidence may decrease when limited clear imagery is available.
• Growth stage and harvest window are model estimates, not field-verified dates.
• The current model covers 8 tillage crop types only; grassland and other land uses are not classified.

Disclaimer: Cube Earth provides decision-support information and should not be used as the sole basis for regulatory or contractual decisions without additional verification. A difference between satellite prediction and LPIS declaration does not, by itself, establish that either is incorrect.

Full methodology: www.cubed-earth.com/methodology.html
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